

Frederick Rajakumar

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EDUCATION

New Jersey Institute of Technology

Newark, NJ

B.S. Computer Science & Applied Physics | GPA: 3.86/4.00

Expected Dec. 2027

Relevant Coursework: DSA, OS, Systems Programming, Computer Networks, Database Systems, Linear Algebra

TECHNICAL SKILLS

Languages: Python, C++, C, TypeScript, JavaScript, SQL, Java, Bash

Web & Cloud: React, Next.js, Node.js, FastAPI, PostgreSQL, REST APIs, Vercel, Railway

Tools: Git, Docker, Linux, GitHub Actions (CI/CD), Playwright

AI & ML: PyTorch, NumPy, LangGraph, Claude API

EXPERIENCE

Undergraduate Researcher – NJIT Department of Computer Science

May 2026 – Present

Advisor: Prof. Xiaoning Ding

Newark, NJ

- Engineered **C++ benchmarks** to evaluate **graph-based ANNS** acceleration on **Intel AMX CPUs** and **NVIDIA GPUs** by scheduling through **Slurm** on NJIT Wulver HPC
- Designed and built a **multithreaded batch controller** for Faiss and Vamana that parallelizes graph traversal to process **106K queries/second** at over **99% Recall@10** on SIFT1M
- Achieved a **3.52× speedup** in vector-distance kernels by batching query distance calculations into **GEMM operations**
- Discovered **memory-latency bottlenecks** on Intel AMX caused by irregular memory access and cache misses

Resident Assistant – New Jersey Institute of Technology

Aug. 2026 – Present

Residence Life

Newark, NJ

- Provide about **20 duty hours/week** of coverage for **60+ residents** while resolving conflicts and documenting incidents
- Co-hosted **2 events** for **80+ residents** by managing budgets, promotion, logistics, and execution

PROJECTS

NJIT Schedule Builder | TypeScript, Next.js, FastAPI, PostgreSQL, Playwright

Summer 2026

<https://njit-schedule-builder-web.vercel.app/scheduler>

- Deployed a full-stack scheduler on Vercel and Railway that generates conflict-free schedules from **6,000+ course sections**
- Designed a backtracking solver that generates up to 25 conflict-free schedules and averaged **2.8 ms per run in local testing**
- Developed a **degree planner** that converts DegreeWorks PDFs into a graduation roadmap using course prerequisites and credit targets

Forge - Multi-Agent C Code Repair | Python, LangGraph, tree-sitter, Docker, cppcheck, clang-tidy

Spring 2026

<https://forge-agentica.streamlit.app/>

- Engineered a **4-agent C-code repair pipeline** using LangGraph to analyze source code and generate patches with an LLM
- Integrated **cppcheck** and **clang-tidy** with tree-sitter call graphs to help the LLM remove duplicate findings and prioritize bugs
- Automated **patch compilation and execution in Docker** using GCC sanitizers and compiler feedback with a limit of 3 attempts per finding

SnapHealth - AI Health Assistant | Python, FastAPI, REST APIs, Claude API, Render

Spring 2026

<https://snaphealth.vercel.app/>

- Delivered a **Python and FastAPI backend** using Claude's vision API to explain symptoms and medication labels from text and images
- Enabled **stateless follow-up chat** using health profiles and recent conversation history to personalize responses
- Handled Claude API failures and malformed JSON by returning **structured fallback responses** when analysis could not be completed

AWARDS & TRAINING

Verizon SMART Campus Challenge Finalist: Received a development stipend to prototype a campus solution (2026)

AWS Cloud Practitioner Essentials: Completed AWS foundational cloud training